



Chapter 7

Chemical Equilibrium

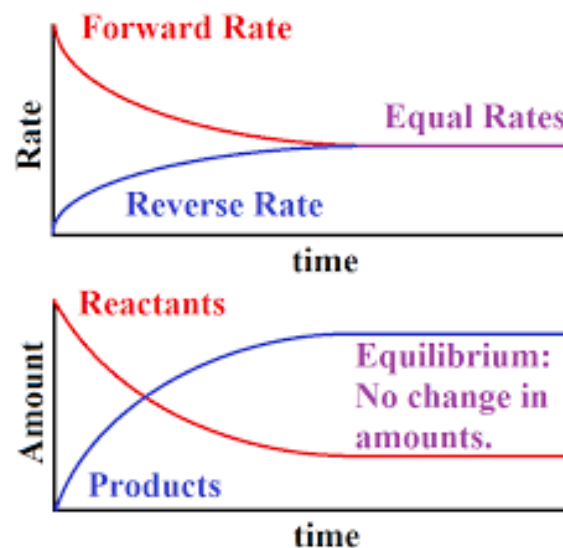


Chemical equilibrium

Chemical reactions move towards a dynamic equilibrium in which both reactants and products are present but have no further tendency to undergo net change.

- The concentrations of reactants and products reach constant values.
- A State of chemical equilibrium is said to have been established, but nothing stopped!
- Chemical equilibrium is a dynamic state

$$\text{Rate}_{\text{forward reaction}} = \text{Rate}_{\text{reverse reaction}}$$





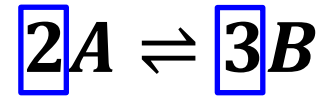
7.1 Spontaneous chemical reactions

1. The Gibbs energy minimum

$$(dG)_{T,p} = -SdT + Vdp + \sum_{i=1}^k \mu_i dn_i$$

At constant T, p ; $W' = 0$

$$(dG)_{T,p} = \sum_{i=1}^k \mu_i dn_i$$



Stoichiometry coefficient

$$\Delta G = \mu_A dn_A + \mu_B dn_B \leq 0$$

$$\Delta G_{mix} = nRT(x_A \ln x_A + x_B \ln x_B)$$

$$dn_A : dn_B$$

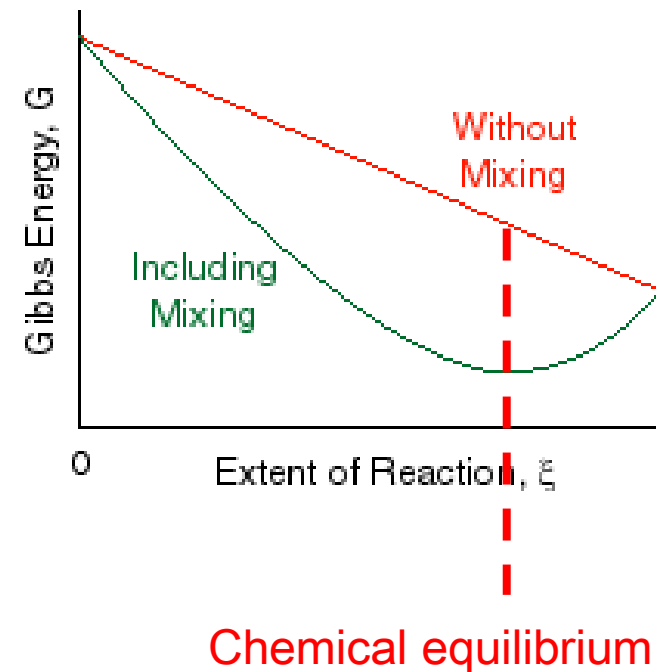
$$-2 : 3$$

ν_i : Stoichiometry number

$$dn_A = \nu_A dx$$

$$dn_B = \nu_B dx$$

$x \rightarrow \text{extent of reaction } \xi$





$$dn_i = \nu_i d\xi$$

$$\Delta n_i = \nu_i \Delta \xi$$

A stoichiometric number is positive for products and negative for reactants.



If $\Delta \xi = 1 \text{ mol}$

	A	B	C	D
ν_i	-2	-1	3	1
Δn_i	-2 mol	- 1 mol	3 mol	1 mol



$$(dG)_{T,p} = \sum_{i=1}^k \mu_i dn_i = \sum_{i=1}^k \mu_i \nu_i d\xi$$

$$\left(\frac{\partial G}{\partial \xi}\right)_{T,p} = \sum_{i=1}^k \nu_i \mu_i = \Delta_r G$$

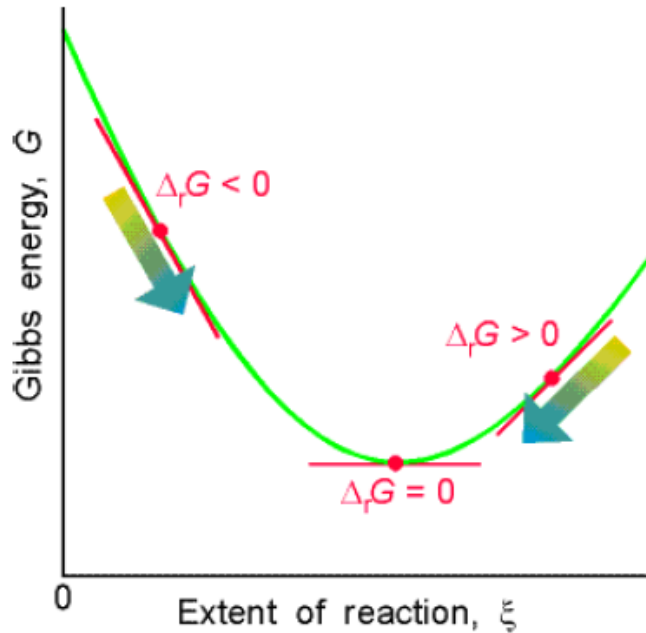
ν_i : Stoichiometry number

ξ : Extent of reaction; The dimensions is amount of substance (moles).

$\Delta_r G$: The reaction Gibbs energy, is defined as the slope of the graph of the Gibbs energy plotted against the extent of reaction, ξ . Dimension (J mol^{-1}).



- $\Delta_r G$



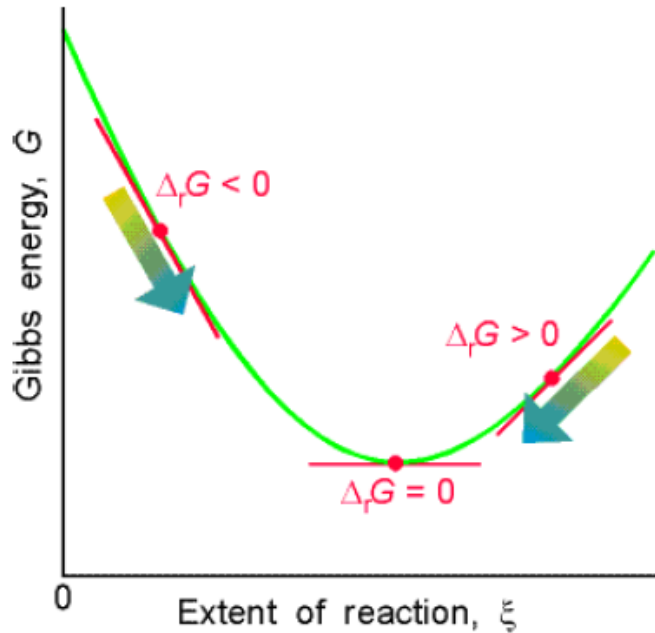
At given T, p

$$\Delta_r G = f(\xi) = f(\text{composition})$$

- Physical meaning: If the reaction advances by $d\xi$, the corresponding changes in Gibbs energy; the slope of a point (the reaction under a certain composition) on the G vs ξ curve.
- When the reaction advances by $d\xi$, $\Delta_r G$ shows no change, which means the reaction has reached its equilibrium.
- When the reaction advances by $d\xi$, $\Delta_r G$ decreases, which means the reaction is a spontaneous process.
- When the reaction advances by $d\xi$, $\Delta_r G$ increases, which means the reverse reaction is a spontaneous process.



The direction of chemical reactions



$$\textcircled{1} \left(\frac{\partial G}{\partial \xi} \right)_{T,p} \quad \textcircled{2} \sum_{i=1}^k \nu_i \mu_i \quad \textcircled{3} \Delta_r G$$

If $\Delta_r G < 0$, the forward reaction is spontaneous; it is called exergonic (work-producing).

If $\Delta_r G > 0$, the reverse reaction is spontaneous; it is called endergonic (work-consuming).

If $\Delta_r G = 0$, the reaction is at equilibrium; it is spontaneous in neither direction, and neither exergonic nor endergonic.



7.2 The description of equilibrium

1. Chemical reaction in gas phase

$$\mu_B = \mu_B^\ominus(T) + RT \ln \frac{p_B}{p^\ominus}$$

$$\Delta_r G = \sum_i \nu_i \mu_i = \sum_i \nu_i \mu_i^\ominus(T) + RT \sum_i \nu_i \ln \frac{p_i}{p^\ominus}$$

$$\underline{\Delta_r G = \Delta_r G^\ominus(T) + RT \ln Q}$$

$$\Delta_r G^\ominus(T) = \sum_i \nu_i \mu_i^\ominus(T) \qquad Q = \prod_i \left(\frac{p_i}{p^\ominus} \right)^{\nu_i}$$

$\Delta_r G^\ominus(T)$: standard reaction Gibbs energy.

Q : reaction quotient.



Reaction quotient



$$Q = \prod_i \left(\frac{p_i}{p^\ominus} \right)^{\nu_i} \quad \longrightarrow \quad Q = \frac{\left(\frac{p_C}{p^\ominus} \right)^3 \left(\frac{p_D}{p^\ominus} \right)^1}{\left(\frac{p_A}{p^\ominus} \right)^2 \left(\frac{p_B}{p^\ominus} \right)^1}$$

- ✓ The reaction quotient measures the relative amounts of products and reactants present during a reaction at a particular point in time.
- ✓ The reaction quotient aids in figuring out which direction a reaction is likely to proceed, given either the pressures or the concentrations of the reactants and the products.



Standard reaction Gibbs energy

Calculation

$$\textcircled{1} \quad \Delta_r G^\ominus = \Delta_r H^\ominus - T \Delta_r S^\ominus$$

$$\Delta_r H^\ominus = \sum_{\text{products}} |v| \Delta_f H^\ominus - \sum_{\text{reactants}} |v| \Delta_f H^\ominus$$

The standard enthalpy of formation, $\Delta_f H^\ominus$: a measure of the energy released or consumed when one mole of a substance is created under standard conditions from its pure elements.

The reference state of an element is its most stable state at the specified temperature and 1 bar.



The Third-Law entropy, standard entropy, $S^{\ominus}(T)$:

On the basis of $S(0\text{K}) = 0$ for all perfect crystals, the entropy of the substance in its standard state at the Temperature T is denoted $S^{\ominus}(T)$

The standard reaction entropy, $\Delta_r S^{\ominus}$: the difference between the molar entropies of the pure, separated products and the pure, separated reactants, all substances being in their standard states at the specified temperature.

$$\Delta_r S^{\ominus} = \sum_{\text{products}} |\nu| S^{\ominus} - \sum_{\text{reactants}} |\nu| S^{\ominus}$$



Standard reaction Gibbs energy

$$\textcircled{2} \quad \Delta_r G^\ominus = \sum_{\text{products}} |\nu| \Delta_f G^\ominus - \sum_{\text{reactants}} |\nu| \Delta_f G^\ominus$$

$\Delta_f G_m^\ominus$: Standard Gibbs energy of formation, the standard reaction Gibbs energy for the formation of compound from its elements in their reference states. Standard free energies of formation of the elements in their reference states are zero.



At equilibrium

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q = 0$$

$$K = \{Q\}_{equilibrium} = \left\{ \prod_i \left(\frac{p_i}{p^\ominus} \right)^{\nu_i} \right\}_{equilibrium}$$

$$\Delta_r G^\ominus = -RT \ln K$$

$$K = \exp \left(-\frac{\Delta_r G^\ominus}{RT} \right)$$

K: Equilibrium constant



$$\Delta_r G = -RT \ln K + RT \ln Q$$

Isothermal equation of reaction

One of the most important equation in chemical thermodynamics: determine the reaction directions

If $Q > K$, $\Delta_r G > 0$

The reaction shift to the LEFT (the reactants are favoured in the equilibrium)

If $Q < K$, $\Delta_r G < 0$

The reaction shift to the RIGHT (the products are favoured in the equilibrium)

If $Q = K$, $\Delta_r G = 0$

The reaction is already at equilibrium.



2. Chemical reaction in solutions

$$\mu_B = \mu_B^\ominus(T) + RT \ln a_B$$

$$\underline{\Delta_r G = \Delta_r G^\ominus(T) + RT \ln Q}$$

$$Q = \prod_i (a_i)^{\nu_i}$$

$$K = \left\{ \prod_i (a_i)^{\nu_i} \right\}_{equilibrium}$$



3. Chemical reaction in multiple phases

$$\underline{\Delta_r G = \Delta_r G^\ominus(T) + RT \ln Q}$$

$$Q = \prod_{B(g)} \left(\frac{f_B}{p^\ominus} \right)^{\nu_B} \cdot \prod_{C(sln)} (a_C)^{\nu_C} \cdot \prod_{D(l)} (a_D)^{\nu_D} \cdot \prod_{E(s)} (a_E)^{\nu_E}$$

For pure liquid and solid

$$a_B = 1$$

$$Q = \prod_{B(g)} \left(\frac{f_B}{p^\ominus} \right)^{\nu_B} \cdot \prod_{C(sln)} (a_C)^{\nu_C}$$

$$K = \{Q\}_{equilibrium}$$



$$K = \left(\frac{p_{\text{CO}_2}}{p^\ominus} \right)_{\text{equilibrium}}$$



4. Thermodynamic equilibrium constant

□ Dimensionless

□ $K = f(T)$ $K = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$

□ Isothermal equation of reaction

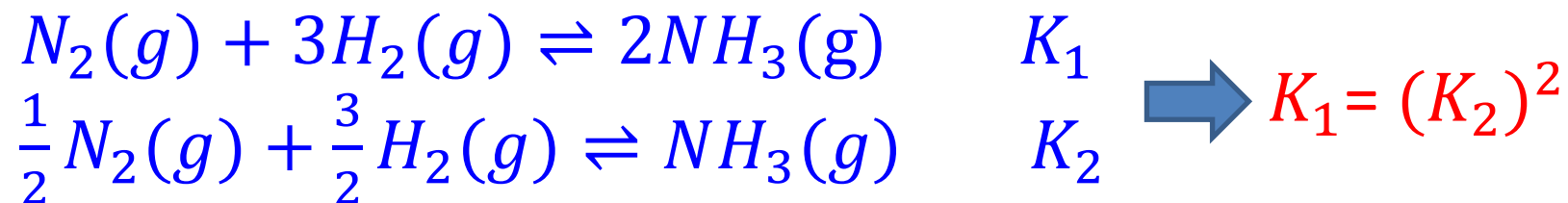
$$\Delta_r G = -RT\ln K + RT\ln Q$$

$$\left\{ \begin{array}{l} \text{If } Q > K, \Delta_r G > 0 \\ \text{If } Q < K, \Delta_r G < 0 \\ \text{If } Q = K, \Delta_r G = 0 \end{array} \right.$$



4. Thermodynamic equilibrium constant

□ Determined by the chemical equations



□ Calculation

$$\textcircled{1} \quad K = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$$

$$\textcircled{2} \quad K = \left\{ \prod_i \left(\frac{p_i}{p^\ominus}\right)^{\nu_i} \right\}_{\text{equilibrium}}$$



Estimating the degree of dissociation at equilibrium

The standard Gibbs energy of reaction for the decomposition



is $+118.08 \text{ kJmol}^{-1}$ at 2300 K. What is the degree of dissociation of H_2O at 2300 K and 1.00 bar?



Method: The equilibrium constant is obtained from the standard Gibbs energy of reaction, so we need to relate the degree of dissociation, α , to K and then to find its numerical value. Proceed by expressing the equilibrium compositions in terms of α , and solve for α in terms of K . Because the standard Gibbs energy of reaction is large and positive, we can anticipate that K will be small, and hence that $\alpha \ll 1$, which opens the way to making approximations to obtain its numerical value.



Answer: The equilibrium constant is obtained from the equation that

$$\ln K = -\frac{\Delta_r G^\ominus}{RT}$$

$$= -\frac{118.08 \text{ kJ mol}^{-1}}{(8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (2300 \text{ K})} = -6.175$$

$$K = 2.08 \times 10^{-3}$$



The equilibrium composition can be expressed in terms of α :

	H_2O	H_2	O_2
Initial amount	n	0	0
Changes to reach equil.	$-\alpha n$	$+\alpha n$	$+\frac{1}{2}\alpha n$
Amount at equili.	$(1-\alpha)n$	αn	$\frac{1}{2}\alpha n$
Mole fraction	$\frac{1-\alpha}{1+\frac{1}{2}\alpha}$	$\frac{\alpha}{1+\frac{1}{2}\alpha}$	$\frac{\frac{1}{2}\alpha}{1+\frac{1}{2}\alpha}$
Partial pressure	$\frac{(1-\alpha)p}{1+\frac{1}{2}\alpha}$	$\frac{\alpha p}{1+\frac{1}{2}\alpha}$	$\frac{\frac{1}{2}\alpha p}{1+\frac{1}{2}\alpha}$



The equilibrium constant is

$$\begin{aligned} K &= \frac{p_{\text{H}_2} p_{\text{O}_2}^{1/2}}{p_{\text{H}_2\text{O}}} \\ &= \frac{\frac{\alpha p}{(1 + \frac{1}{2} \alpha)} \left(\frac{\frac{1}{2} \alpha p}{(1 + \frac{1}{2} \alpha)} \right)^{1/2}}{\frac{(1 - \alpha) p}{(1 + \frac{1}{2} \alpha)}} \\ &= \frac{\alpha^{2/3} p^{1/2}}{(1 - \alpha)(2 + \alpha)^{1/2}} \end{aligned}$$



$$K = \frac{\alpha^{2/3} p^{1/2}}{(1 - \alpha)(2 + \alpha)^{1/2}}$$

In this expression, p in place of $p/p^\ominus$ to keep the notation simple. Now make the approximation that $\alpha \ll 1$:

$$K \approx \frac{\alpha^{2/3} p^{1/2}}{\sqrt{2}}$$

Under the stated conditions, $p=1.00$ (that is, $p/p^\ominus \equiv 1.00$), so

$$\alpha \approx (\sqrt{2}K)^{2/3} = 0.0205$$

That is, about 2 per cent of the water has decomposed.



7.2 The response of equilibria to the conditions

Le Chatelier's principle

A system at equilibrium, when subjected to a disturbance, responds in a way that tends to minimize the effect of the disturbance.

There are several ways to stress an equilibrium. One way is to add or remove a product or a reactant in a chemical reaction at equilibrium. When additional reactant is added, the equilibrium shifts to reduce this stress: it makes more product. When additional product is added, the equilibrium shifts to reactants to reduce the stress. If reactant or product is removed, the equilibrium shifts to make more reactant or product, respectively, to make up for the loss.



1. How equilibria respond to pressure (gas phase reactions)

$$K = f(T) \quad K = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$$

The conversion rate α of species A in a reaction is equal to the number of moles of A reacted per mole of A fed:



initial	n	0
reacted	$-\alpha n$	$+2\alpha n$
Final in equilibrium	$(1-\alpha)n$	$+2\alpha n$

$$(x_A)_{eq} = \frac{(1-\alpha)n}{(1-\alpha)n + 2\alpha n} = \frac{1-\alpha}{1+\alpha}$$

$$(x_B)_{eq} = \frac{2\alpha}{1+\alpha}$$

$$K = \frac{p_B^2}{p_A} = \frac{x_B^2 p}{x_A} = \frac{4\alpha^2 p}{1-\alpha^2} \quad \Rightarrow \quad \alpha = \left(\frac{1}{1+4p/K}\right)^{1/2} \quad \left\{ \begin{array}{l} p \uparrow \\ \alpha \downarrow \end{array} \right.$$



- $K = f(T)$

- $K = \prod_i \left(\frac{p_i}{p^\ominus}\right)^{\nu_i} = \prod_i \left(\frac{x_i p}{p^\ominus}\right)^{\nu_i} = \prod_i x_i^{\nu_i} \cdot \prod_i \left(\frac{p}{p^\ominus}\right)^{\nu_i}$

$$K = K_x \cdot \left(\frac{p}{p^\ominus}\right)^{\sum \nu_i}$$



Condition I

When a reaction at equilibrium is compressed: $p \uparrow$

$$K = K_x \cdot \left(\frac{p}{p^\ominus}\right)^{\sum \nu_i}$$

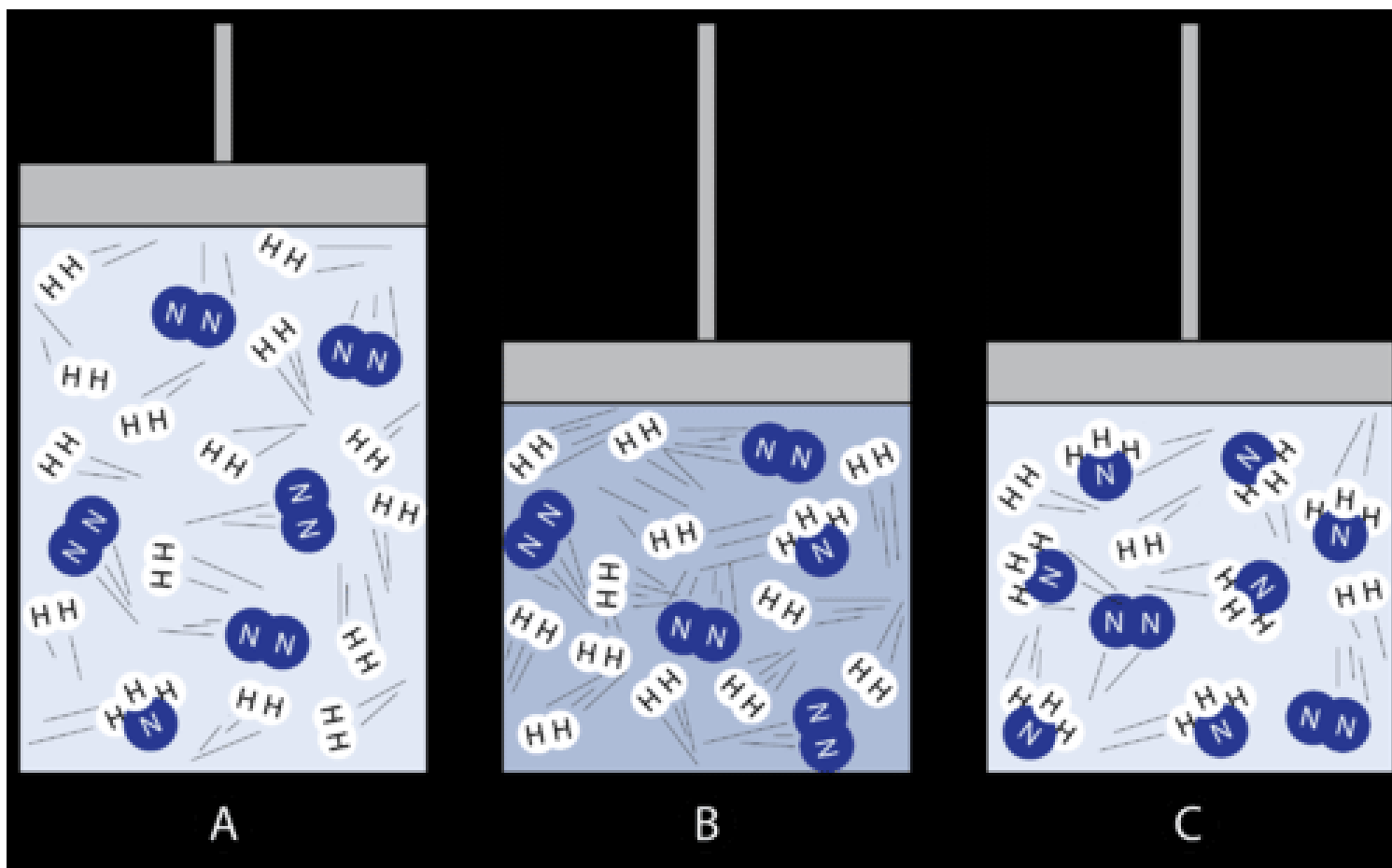
$$\because K = \text{constant}; p \uparrow$$

If $\sum \nu_i > 0$; $K_x \downarrow$ *Shift Left*

If $\sum \nu_i < 0$; $K_x \uparrow$ *Shift Right*

If $\sum \nu_i = 0$; $K_x \rightarrow$ *keep equilibrium*

Increasing the pressure on a equilibrium system will result in the equilibrium position shifting to reduce the pressure, that is, to the side that has the least number of gas particles





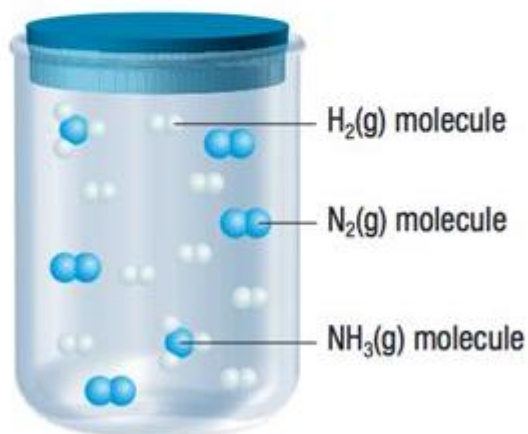
Condition II

The pressure within a reaction vessel can be increased by injecting an inert gas into it. However, as long as the gases are perfect, this addition of gas leaves all the partial pressures of the reacting gases unchanged: the partial pressure of a perfect gas is the pressure it would exert if it were alone in the container, so the presence of another gas has no effect.



- **Inert Gas** – adding an inert gas will increase the pressure thus increasing number of collisions but since it is inert the collisions will not cause reactions leaving the equilibrium intact

p. 445





2. The response of equilibria to temperature

$$\ln K = -\frac{\Delta_r G^\ominus}{RT} \quad \longrightarrow \quad \frac{d \ln K}{dT} = -\frac{1}{R} \frac{d(\Delta_r G^\ominus / T)}{dT}$$

Gibbs-Helmholtz equation

$$\frac{d(\Delta_r G^\ominus / T)}{dT} = -\frac{\Delta_r H^\ominus}{T^2}$$

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

Review Chapter 3

The van't Hoff equation

$$\frac{d(1/T)}{dT} = -\frac{1}{T^2}$$

$$\therefore dT = -T^2 d(1/T)$$

$$\frac{d \ln K}{d(1/T)} = -\frac{\Delta_r H^\ominus}{R}$$

The van't Hoff equation



$$\frac{d \ln K}{d(1/T)} = - \frac{\Delta_r H^\ominus}{R}$$

If $\Delta_r H^\ominus < 0$; *Exothermic reaction*
 $T \uparrow \quad K \downarrow \quad \text{Shift Left}$

If $\Delta_r H^\ominus > 0$; *Endothermic reaction*
 $T \uparrow \quad K \uparrow \quad \text{Shift Right}$



The value of K at different temperatures

$$\frac{d \ln K}{d(1/T)} = - \frac{\Delta_r H^\ominus}{R}$$

$$\ln K_2 - \ln K_1 = - \frac{1}{R} \int_{1/T_1}^{1/T_2} \Delta_r H^\ominus d(1/T)$$

When $\Delta_r H^\ominus$ varies only slightly with temperature over the temperature range of interest

$$\ln K_2 - \ln K_1 = - \frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Estimate the equilibrium constant for the synthesis of ammonia at 500 K from its value at 298 K (6.0×10^5 for the reaction) as written in the following:





Answer:



because $\Delta_r H^\ominus = 2\Delta_f H^\ominus(\text{NH}_3, \text{g})$ and assume that its value is constant over the range of T :

$$\begin{aligned} \ln K_2 - \ln K_1 &= -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \\ &= \ln(6.0 \times 10^5) - \frac{-92.2 \text{ kJ mol}^{-1}}{8.314 \text{ J K}^{-1} \text{ mol}^{-1}} \left(\frac{1}{500 \text{ K}} - \frac{1}{298 \text{ K}} \right) \\ &= -1.73 \end{aligned}$$

It follows that $K_2 = 0.18$



Summary

1. Extent of reaction ξ

$$dn_i = \nu_i d\xi$$

$$d\xi = \frac{dn_i}{\nu_i}$$

2. The direction of chemical reactions

T, p constant, $W'=0$

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{T,p} = \sum_{i=1}^k \nu_i \mu_i \leq 0$$



3. Isothermal equation of reaction

$$\Delta_r G = \Delta_r G^\ominus(T) + RT \ln Q \quad Q = \prod_i (a_i)^{\nu_i}$$

$$\Delta_r G^\ominus = -RT \ln K \quad K = \left\{ \prod_i (a_i)^{\nu_i} \right\}_{\text{equilibrium}}$$

$$\Delta_r G = -RT \ln K + RT \ln Q \quad \left\{ \begin{array}{l} \text{If } Q > K, \Delta_r G > 0 \\ \text{If } Q < K, \Delta_r G < 0 \\ \text{If } Q = K, \Delta_r G = 0 \end{array} \right.$$



4. Thermodynamic equilibrium constant

$$K = \left\{ \prod_i (a_i)^{\nu_i} \right\}_{equilibrium}$$

For perfect gas

$$K = \left\{ \prod_i \left(\frac{p_i}{p^\ominus} \right)^{\nu_i} \right\}_{equilibrium}$$

$$K = K_x \cdot \left(\frac{p}{p^\ominus} \right)^{\sum \nu_i}$$



For solutions

$$K = \left\{ \prod_i (a_i)^{\nu_i} \right\}_{equilibrium}$$

For multiple phases

$$Q = \prod_{B(g)} \left(\frac{p_i}{p^\ominus} \right)^{\nu_i} \cdot \prod_{B(sln)} (a_i)^{\nu_i}$$

$$K = \{Q\}_{equilibrium}$$



- Chemical potential

$$\mu = \mu^{\ominus} + RT \ln a$$



$$a = 1$$

a 无量纲

$$\mu = \mu^{\ominus}$$

所以standard state首先是 $a=1$ 的状态



纯气体

$$\mu = \mu^{\ominus}(T) + RT \ln \frac{p}{p^{\ominus}}$$

standard state: $\frac{p}{p^{\ominus}} = 1$

pure gas, $T, p = p^{\ominus} = 1 \text{ bar}$

混合气体

$$\mu_B = \mu_B^{\ominus}(T) + RT \ln \frac{p_B}{p^{\ominus}}$$

standard state: $\frac{p_B}{p^{\ominus}} = 1$

pure gas B, $T, p = p^{\ominus} = 1 \text{ bar}$



理想混合气体中某种气体的分压等于这种气体单独占据整个体积时的压力



溶液以溶剂为观测对象

$$\mu_A = \mu_A^\ominus + RT \ln a_A = \mu_A^\ominus + RT \ln \gamma_A x_A$$

溶液越稀薄，溶剂的比例越高，溶液越接近理想溶液，符合Raoult's law

$$\lim_{x_A \rightarrow 1} \gamma_A = 1$$

standard state:

$$a_A = 1$$

pure liquid A, T, p[⊖]



溶液以溶质为观测对象

$$\mu_B = \mu_B^\ominus + RT \ln a_B = \mu_B^\ominus + RT \ln \gamma_B \frac{b_B}{b^\ominus}$$

溶液越稀薄，溶液越接近理想稀薄溶液，符合Henry's law

$$\lim_{b_B \rightarrow 0} \gamma_B = 1$$

standard state:

$$a_B = 1$$

$$\gamma_B = 1 \quad b_B = b^\ominus = 1 \text{ mol kg}^{-1}$$

$$b_B = 1 \text{ mol kg}^{-1}, \quad T, p \text{ obeys Henry's law,}$$

Hypothetical state

因此 $a=1$ 的状态是远离Henry's law的状态，因此我们假想在 $a=1$ 的状态下仍然符合Henry's law



$$\mu_i = \mu_i^\ominus + RT \ln a_i$$

- 化学反应是随着反应进行，体系内组成一直在动态变化的复杂体系。
- The chemical reaction is a complex system with dynamic changes in the composition as the reaction progresses.

$$(dG)_{T,p} = \sum_{i=1}^k \mu_i dn_i$$

- 但各个组成的变化不是没有内在联系的随意变化，而是有化学计量关系的变化
- However, the change of each composition is not an arbitrarily change without intrinsic connection but in the stoichiometric relationship.

$$dn_i = \nu_i d\xi \quad d\xi = \frac{dn_i}{\nu_i}$$

ν_i : Stoichiometry number

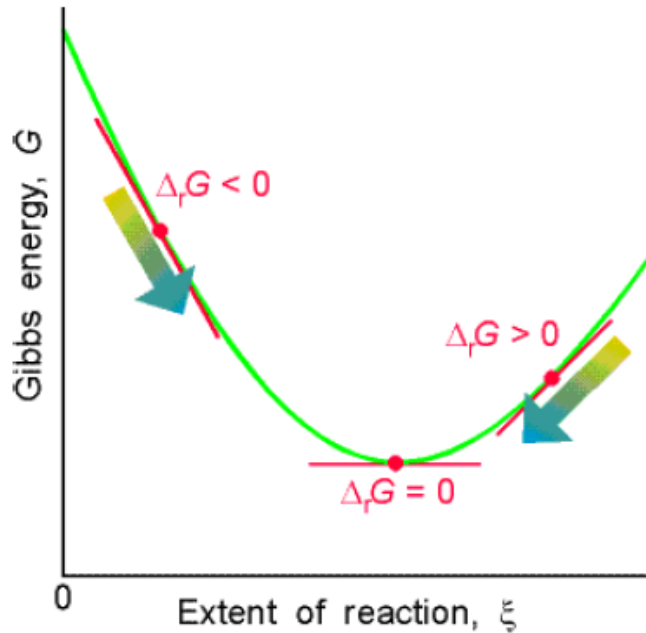
ξ : Extent of reaction (moles). 反应进行程度的表征

- The reaction Gibbs energy (J/mol)

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{T,p} = \sum_{i=1}^k \nu_i \mu_i \leq 0 \quad \Rightarrow \quad \Delta_r G = \Delta_r G^\ominus(T) + RT \ln Q$$



- $\Delta_r G$



At given T, p

$$\Delta_r G = f(\xi) = f(\text{composition})$$

- Physical meaning: If the reaction advances by $d\xi$, the corresponding changes in Gibbs energy; the slope of a point (the reaction under a certain composition) on the G vs ξ curve.
- When the reaction advances by $d\xi$, $\Delta_r G$ shows no change, which means the reaction has reached its equilibrium.
- When the reaction advances by $d\xi$, $\Delta_r G$ decreases, which means the reaction is a spontaneous process.
- When the reaction advances by $d\xi$, $\Delta_r G$ increases, which means the reverse reaction is a spontaneous process.



- $\Delta_r G^\ominus$

- ✓ 反应物和产物都在标准状态下 All the reactant and products are under standard conditions
- ✓ 温度的函数 $\Delta_r G^\ominus = f(T)$
- ✓ 算法

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$

$$\Delta_r G^\ominus = \sum_{\text{products}} \nu \Delta_f G_m^\ominus - \sum_{\text{reactants}} \nu \Delta_f G_m^\ominus$$

$$\Delta_r G^\ominus = -RT \ln K$$



• Q

- ✓ The reaction quotient measures the relative amounts of products and reactants present during a reaction at a particular point in time.

$$Q = \prod_{B(g)} \left(\frac{f_B}{p^\ominus} \right)^{\nu_B} \cdot \prod_{C(sln)} (a_C)^{\nu_C}$$



连乘号



• K Equilibrium constant

✓ $K = \{Q\}_{equilibrium}$

✓ Dimensionless 无量纲

✓ $K = f(T)$ $K = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$ 温度的函数

✓ 可以判断反应方向: isothermal equation of reaction

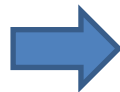
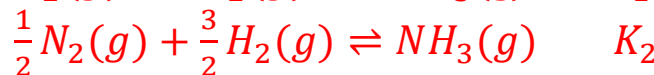
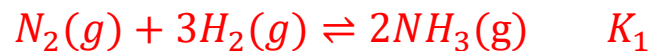
$$\Delta_r G = -RT \ln K + RT \ln Q$$

If $Q > K$, $\Delta_r G > 0$

If $Q < K$, $\Delta_r G < 0$

If $Q = K$, $\Delta_r G = 0$

✓ Determined by the chemical equations



$$K_1 = (K_2)^2$$

✓ 计算

$$\textcircled{1} \quad K = \exp\left(-\frac{\Delta_r G^\ominus}{RT}\right)$$

$$\textcircled{2} \quad K = \left\{ \prod_i \left(\frac{p_i}{p^\ominus} \right)^{v_i} \right\}_{equilibrium}$$



Le Chatelier's principle---equilibrium shift

Le Chatelier's principle

A system at equilibrium, when subjected to a disturbance, responds in a way that tends to minimize the effect of the disturbance.



Temperature

$$\frac{d \ln K}{d(1/T)} = - \frac{\Delta_r H^\ominus}{R}$$

If $\Delta_r H^\ominus > 0$; *Exothermic reaction*
 $T \uparrow \quad K \downarrow \quad \text{Shift Left}$

If $\Delta_r H^\ominus < 0$; *Endothermic reaction*
 $T \uparrow \quad K \uparrow \quad \text{Shift Right}$



Pressure

Condition I

When a reaction at equilibrium is compressed: $p \uparrow$

$$K = \prod_i \left(\frac{p_i}{p^\ominus}\right)^{\nu_i} = \prod_i \left(\frac{x_i p}{p^\ominus}\right)^{\nu_i} = \prod_i x_i^{\nu_i} \cdot \prod_i \left(\frac{p}{p^\ominus}\right)^{\nu_i}$$

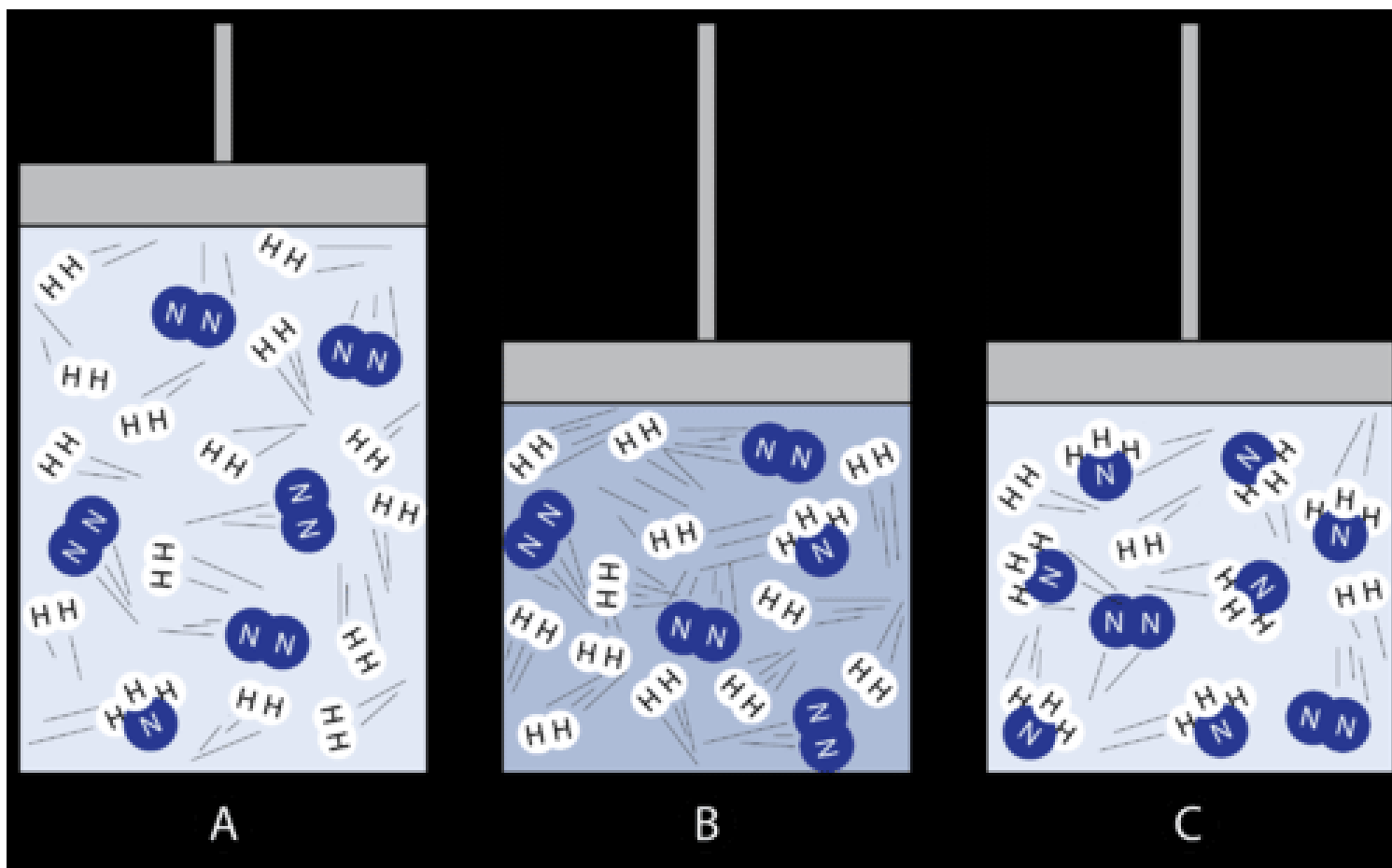
$$K = K_x \cdot \left(\frac{p}{p^\ominus}\right)^{\sum \nu_i}$$

$$\because K = \text{constant}; p \uparrow$$

If $\sum \nu_i > 0$; $K_x \downarrow$ *Shift Left*

If $\sum \nu_i < 0$; $K_x \uparrow$ *Shift Right*

If $\sum \nu_i = 0$; $K_x \rightarrow$ *keep equilibrium*





Condition II

The pressure within a reaction vessel can be increased by injecting an inert gas into it. However, as long as the gases are perfect, this addition of gas leaves all the partial pressures of the reacting gases unchanged: the partial pressure of a perfect gas is the pressure it would exert if it were alone in the container, so the presence of another gas has no effect.



- **Inert Gas** – adding an inert gas will increase the pressure thus increasing number of collisions but since it is inert the collisions will not cause reactions leaving the equilibrium intact

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